KIRUBIA A

Data Analyst · Machine Learning · Power BI & Tableau

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PROFESSIONAL SUMMARY

Results-driven Data Analyst with a B.Tech in AI & Data Science (CGPA 8.2) and hands-on experience building end-to-end ML pipelines, ETL workflows, interactive dashboards, and business-impact analyses on datasets exceeding 150,000 records. Delivered a 12% accuracy improvement in production models and translated complex findings into executive-ready insights. Proficient in Python, SQL, Power BI, and Tableau with exposure to GCP (BigQuery) for cloud-based data processing — ready to drive decisions from day one.

TECHNICAL SKILLS

Languages	Python (Pandas, NumPy), SQL (MySQL — joins, window functions, CTEs), Excel (Pivot Tables, XLOOKUP)
Visualisation	Power BI (DAX, KPI cards, slicers), Tableau, Matplotlib, Seaborn, Plotly, Streamlit
ML & AI	Scikit-learn, XGBoost, LightGBM, Random Forest, Logistic Regression, K-Means, SHAP, SMOTE
NLP	NLTK, spaCy, TF-IDF, VADER Sentiment Analysis
Cloud & Data	GCP (BigQuery, Colab), MLflow, Git, GitHub, Jupyter Notebook, VS Code
Core Concepts	ETL Pipelines, Data Modeling, EDA, Feature Engineering, Statistical Analysis, Hypothesis Testing, Data Storytelling, Business Intelligence

PROFESSIONAL EXPERIENCE

Data Science Intern | Gradtwin Services — *Sept 2024 – Nov 2024*

- Engineered end-to-end ETL pipeline with feature engineering (label encoding, one-hot encoding, normalisation) and hyperparameter tuning via GridSearchCV — boosting production model accuracy by 12%.
- Built and benchmarked ML models (KNN, Random Forest) using 5-fold cross-validation; selected best-fit model using F1-score, ROC-AUC, precision, and recall — reducing misclassification rate.
- Uncovered actionable trends and outliers in real-world datasets via exploratory analysis using Pandas, Matplotlib, and Seaborn; findings directly shaped client-facing recommendations.
- Delivered 4+ stakeholder-ready business intelligence reports translating technical model outputs into clear narratives with supporting visualisations.

KEY PROJECTS

Nykaa Beauty Brand Review Analysis — *Python · Pandas · SQL · Matplotlib · Seaborn · VADER · Power BI*

- Built an end-to-end ETL pipeline processing and cleaning 61,284 customer reviews across 11 brands — covering data ingestion, SQL querying, EDA, and VADER sentiment scoring.
- Surfaced that Kay Beauty led the category with a 4.69 avg rating and highest 5-star ratio — insight directly actionable for brand partnership strategy.
- Designed a 9-visual interactive Power BI dashboard featuring brand slicers, KPI cards, and trend charts; reduced ad-hoc reporting time by ~30% for stakeholders.

Global AI Adoption & Workforce Impact — *Python · XGBoost · LightGBM · K-Means · SHAP · MLflow · Streamlit · SMOTE*

- Designed and deployed a full ML pipeline on 150,000+ records spanning 20+ industries — XGBoost multi-class classifier with SHAP explainability, K-Means segmentation, and MLflow experiment tracking.
- Resolved severe class imbalance with SMOTE; tracked 12+ experiment runs via MLflow, accelerating data modeling iteration cycle.
- Deployed a real-time Streamlit dashboard with interactive filters, enabling non-technical stakeholders to explore model outputs independently.

Sentiment Analysis — Product Reviews — *Python · NLTK · spaCy · TF-IDF · Logistic Regression · Scikit-learn*

- Built end-to-end NLP pipeline: text cleaning → tokenisation → stopword removal → TF-IDF vectorisation → Logistic Regression classifier.
- Evaluated model rigorously using precision, recall, F1-score, and confusion matrix, achieving production-grade classification performance.

Sales Data Analysis & KPI Dashboard — *Python · SQL · Tableau · Excel · Pandas · Seaborn*

- Analysed 9,800+ sales records using SQL and Python; identified excessive discounting as a net revenue drain — delivered corrective pricing recommendations to stakeholders.
- Built Tableau KPI dashboards visualising revenue trends, deal velocity, and discount impact — adopted as the team's primary business intelligence reporting tool.

Cirrhosis Disease Stage Prediction — *Python · Scikit-learn · Random Forest · Logistic Regression · SMOTE*

- Trained multi-class classifier on clinical data; Random Forest outperformed baseline by 18% — selected as production model.
- Applied SMOTE for class imbalance; evaluated across all stages using accuracy, F1-score, and AUC-ROC.

EDUCATION

B.Tech — Artificial Intelligence & Data Science | Saveetha School of Engineering, Chennai — *2021 – 2025*

CGPA: 8.2 / 10 · Relevant Coursework: Machine Learning, Deep Learning, NLP, Statistics, Database Systems, Data Visualisation

CERTIFICATIONS

- Accenture Data Analytics & Visualisation Job Simulation — Forage | Aug 2024
- Oracle Cloud Infrastructure GenAI Certified Professional — Oracle | Jul 2024
- Oracle Cloud Infrastructure AI Foundations Associate — Oracle | Aug 2024